

5 CORE 200 Reading | Module 5 | Interventions

Contents

5 CORE 200 Reading | Module 5 | Interventions

5.0.1 Module 5 | Introduction 1

5.0.2 Concepts and Skills Covered in Module 5 2

5.1 The Idea of Intervention 2

5.1.1 Module 5 | Interventions as Actions 2

5.1.2 Module 5 | Ideal Interventions 3

5.2 Interventions and Causal Relationships 4

5.2.1 Module 5 | Values and Relationships 4

5.2.2 Module 5 | Intervening on Causes 4

5.2.3 Module 5 | Intervening on Effects 5

5.2.4 Module 5 | Natural and Experimental Causal Systems 6

5.2.5 Module 5 | Exogenous and Endogenous Variables and Causal Structure 7

5.2.6 Module 5 | Interventions on Cyclic Causal Systems 8

5.2.7 Module 5 | Case Studies 8

5.3 Summary 11

5.3.1 Module 5 | Module Summary 11

5.3.2 Concepts and Skills Covered in Module 5 11

5.0.1 Module 5 | Introduction

Our account of causation involves causal assignments. A crucial feature of a causal assignment is that the values of variables are assigned, and not observed. The difference between assigning and observing is at the root of causation, and is the topic of this module.

For example, consider two variables that relate to traveling on an airplane: Seat Belt Sign [lit, unlit], Turbulence [yes, no]. It is quite often the case that the seat belt light goes on a few moments before turbulence. If we build a response structure for Turbulence, however, we must consider two causal assignments:

Response Structure For Turbulence		
Assignment	Seat Belt Sign	Turbulence
1	Lit	??
2	Unlit	??

Assuming we take causal assignment 1 seriously, then the question is: if we *intervene* at any arbitrary time to light up the Seat Belt signs, will Turbulence follow? The answer is clearly no, as the sign itself has no influence on whether the plane will hit turbulence. Turbulence often follows shortly after the seat-belt sign is lit up because the pilot puts it on in response to a weather report telling him or her that the plane is about to hit some turbulence. During normal flights we *observe* the sign light up, and then *observe* that turbulence follows. To claim that the Seat Belt Sign being lit is a “cause” of turbulence, we should believe that arbitrary interventions that force the sign to light up change the chances of experiencing Turbulence soon thereafter.

In this module, we examine what an *intervention* is, and describe the kind that we consider ideal. Ideal interventions are the foundation of causal claims and mark the surest path to causal discovery.

The essential ideas are these:

- An *ideal intervention* directly fixes the value of a variable within a causal system.
- Ideally intervening to change the value of a cause does not change the relationship between the cause and its effects.
- Ideally intervening to change the value of an effect does change the relationship between the effect and its causes.

Because ideal interventions can change the relationship between an effect and its causes, we distinguish between the causal system as it occurs *before* an intervention and the system as it occurs after the intervention. We call the pre-intervention system the “natural” or “pre-manipulated” system, and the post-intervention system the “experimental,” “post-manipulation,” or “manipulated” system.

When you finish this module, you should understand what an ideal intervention is and how it changes a causal system. You should be able to take 1) a causal graph that describes a “natural” or “pre-manipulated” system and 2) an intervention that we might perform on that system, and produce the “experimental,” or “manipulated” causal graph that would result.

The module stresses six basic skills and/or concepts:

5.0.2 Concepts and Skills Covered in Module 5

1. Intervention as Action
2. Ideal Interventions
3. Interventions and Causal Relationships
4. Pre-Manipulation vs. Post-Manipulation Graphs
5. Exogenous vs. Endogenous Variables
6. Learning Causal Effects from Interventions

5.1 The Idea of Intervention

5.1.1 Module 5 | Interventions as Actions

Sometimes a variable changes from naturally occurring causes, and sometimes we *act* to change it. For example, the noise level in my study sometimes changes when my children come in or go out, or when the neighbor finishes mowing his lawn. Sometimes, however, I act to make it quiet by kicking my kids out and shutting the window.

Another example is the amount of money in my checking account, which can be categorized as low or high, depending, let's say, on whether it has under \$2000 or over \$2000. The level naturally varies quite a lot during the course of a month, depending on the deposits or debits that come in daily. Suppose the most important factor for me is whether my paycheck has been deposited recently (within 3 days) or not. So consider the causal system: {CHECK DEPOSITED RECENTLY, CHECKING ACCOUNT LEVEL}, which we will represent with this causal graph:



There are other factors that will affect my checking account, e.g., whether I buy an airplane ticket for my whole family that month or not, but I won't include any of these in my system.

At most times during the month, I am capable of intervening from outside this system to change the level of my checking account. I can:

- go into the bank and withdraw my entire account in cash (the total withdrawal intervention), or
- deposit \$2,500 in cash that I have stashed in my attic (the \$2500 deposit intervention).

An intervention is an *action*. Deciding to withdraw all the money from my account is not an action, it is a decision. Going to the bank and filling out the appropriate paperwork for a total withdrawal of my account is an intervention. In addition, an intervention is not just an observation. Looking at my account balance and seeing that it is low is an observation. Actually withdrawing all my money is an intervention.

An intervention does not necessarily change the value of the variable. For example, if I already have over \$2000 in my checking account, depositing \$2500 in cash will not change the value of the variable Checking Account – the value will

still be High. The act is still an ideal intervention, because it *determines* the value of the variable, regardless of its value before the intervention.

In practical cases we usually restrict our attention to interventions that are actions actually under our direct control. For example, “getting my friend to stop smoking” is not a practical intervention per se, because it doesn’t clearly specify an action under your control. “Taking my friend to dinner and lecturing him on the perils of smoking” is an action under your control. “Reducing crime” is not an action under the government’s actual control; it is an outcome that candidates hope you believe they can indirectly control. “Increasing the FBI’s budget by \$100 million” is an action that the government can actually perform, as is “increasing the budget for crime prevention programs.”

Even if a hypothetical intervention is not practically possible, we can still often sensibly reason about it. For example, it is not practically possible to give every resident of Iowa a million dollars, but it is perfectly clear what it is we are considering doing, and we can reason about the economic consequences in Iowa were we to actually execute this intervention. In contrast, some descriptions of possible interventions are too vague to even consider hypothetically within a scientific framework. For example: “making all relationships loving,” or “creating equal opportunity for all.”

5.1.2 Module 5 | Ideal Interventions

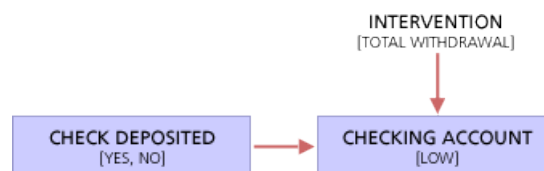
In this section, we describe the simplest sorts of interventions, which we will call ideal interventions.

Ideal Intervention An *ideal intervention* is an action which

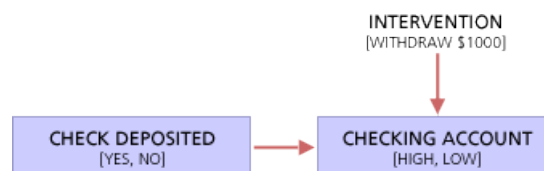
1. directly targets exactly one variable in the causal system, and
2. is completely effective in determining the value of that variable, and
3. is itself exogenous, or uncaused.

For example, the total withdrawal intervention only directly targets the level of my checking account and does not influence whether my paycheck is deposited recently or not. It is also completely effective in determining that the level of my checking account is “low.” Even if my checking account was already low, this act would still count as an ideal intervention, because it *determines* the level of my checking account.

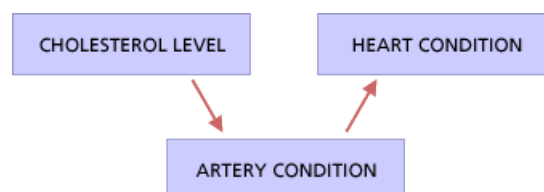
An intervention is exogenous just in case nothing else in the causal system under study is influencing it. If I only did the total withdrawal intervention after learning that a check had just been deposited, the exogeneity assumption would be violated. For an intervention to be ideal, it must come from outside the system under study.



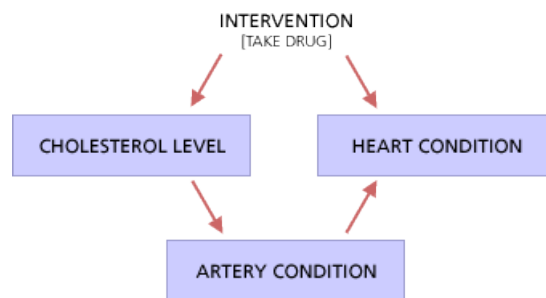
Contrast this with an intervention that withdraws \$1,000 from my account (the \$1,000 intervention). Although this does directly target only the checking account level, it is not completely effective in determining whether my checking account is high (above \$2000), or low (below \$2,000). It will certainly change the chances that my account is high or low, but not determine it.



Other interventions might intend to target only one variable, but actually directly affect others in the system. Consider the following causal system involving cholesterol, the condition of your arteries, and the condition of your heart:



Suppose your doctor decides to intervene to reduce your cholesterol level, intending to improve your heart condition indirectly. Suppose she prescribes a drug that reduces your body’s production of cholesterol, has no direct effect on the condition of your arteries, but that independently induces a chronic arrhythmia which seriously damages your heart.



Although the intervention was intended to target only cholesterol, we say that it had a “fat hand” because it inadvertently also directly affected the condition of your heart.

If, on the other hand, the drug does manage to target only your cholesterol, then your heart condition would be improved. It would seem then, that the intervention, even if it worked the way we wanted, would still have an effect on more than one variable. Wouldn’t it still be a “fat hand”? The answer is no, because in this second case the drug only indirectly affects your heart condition. An ideal intervention must not affect more than one variable directly; otherwise it’s a “fat hand.”

Not all interventions are ideal, and the theory of interventions and causal graphs in no way depends on the interventions being ideal. For this module and those that follow, however, we will assume that the interventions we consider are ideal. That is, we will assume that interventions are actions that directly and effectively change the value of one and only one variable in a causal system.

5.2 Interventions and Causal Relationships

5.2.1 Module 5 | Values and Relationships

Interventions directly change the value of a variable in a causal system, but they can also change the causal *relationships* themselves. In general, interventions that change the value of a cause leave intact the relationship between the cause and its effects, but interventions that change the value of an effect change the relationship between the effect and its causes. In this section we explore how this happens and its consequences for causal reasoning.

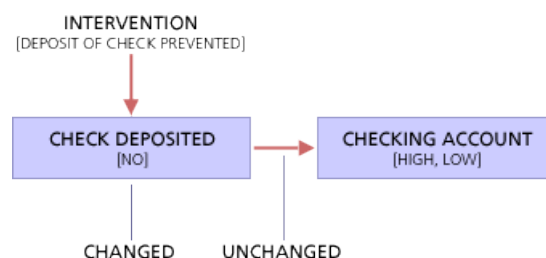
5.2.2 Module 5 | Intervening on Causes

🔗 Example (A) Example 1

Consider again the simple causal system involving my checking account:



In this system, there is a causal relationship posited to exist between my paycheck being deposited and the state of my checking account. Suppose we intervene to prevent my paycheck from being deposited, which we model with a new variable coming in from the outside:

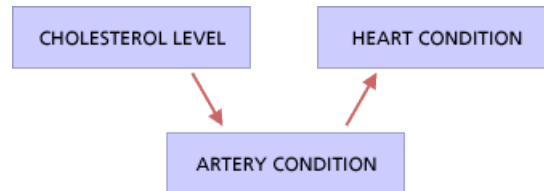


The intervention changes the *value* of the paycheck variable, but it does not affect the *relationship* between the Paycheck variable and the Checking Account variable. When my paycheck is not deposited recently, that will still have an effect on whether my checking account is high or low.

In general, if there is no cycle between a cause and its effect, then intervening to change the value of a cause does not change the relationship between the cause and its effects.

Example (B) Example 2

Consider again the simple causal system involving cholesterol and your heart:



In this system, there is a causal relationship posited to exist between cholesterol and artery condition.

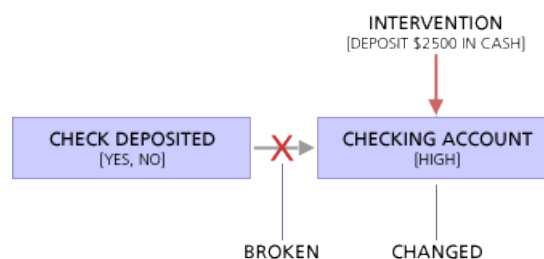
5.2.3 Module 5 | Intervening on Effects

Example (C) Example 1

An ideal intervention on an effect does change the relationship between the effect and its causes. Consider again the simple causal system involving my checking account:



Whereas before we intervened on my paycheck, which is the cause, now let's intervene on the effect (the checking account) and see what happens. Consider the \$2500 deposit intervention (I deposit \$2,500 in cash). Again, we model the intervention with a new variable coming in from the outside:

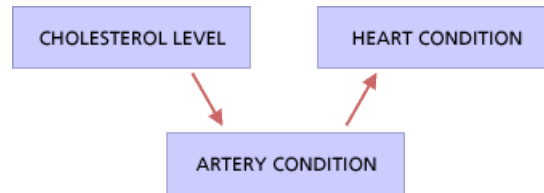


The intervention changes the value of the Checking Account variable, but it also severs the relationship between the Paycheck variable and the Checking Account variable. The Paycheck deposit no longer has any influence on whether my checking account is high. The deposit \$2500 in cash intervention took over all influence, and in so doing changed the *relationship* between the effect and its causes.

In general, if there is no cycle between a cause and its effect, then ideally intervening to change the value of the effect *breaks the relationship between the effect and its causes*.

Example (D) Example 2

Consider again the simple causal system involving cholesterol and your heart:

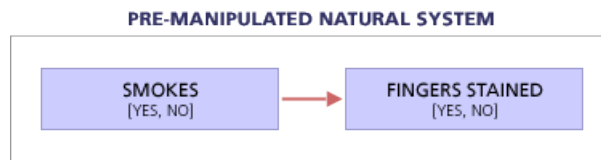


In this system, there is a causal relationship posited to exist between cholesterol and artery condition.

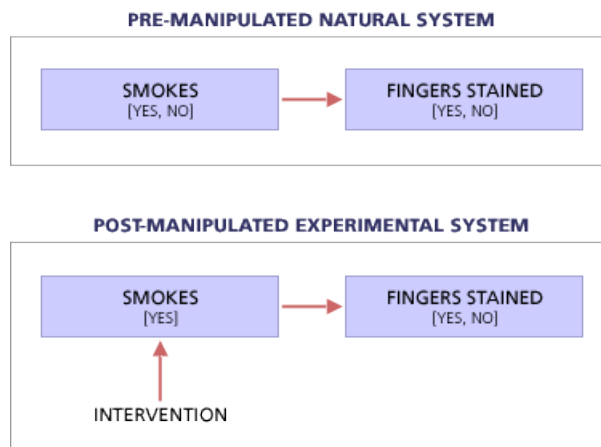
5.2.4 Module 5 | Natural and Experimental Causal Systems

Because interventions can change the relationship between an effect and its causes, to keep things clear we distinguish between the causal system as it occurs *before* an intervention and the system as it occurs *after* the intervention. We call the pre-intervention system the “natural” or “pre-manipulated” system, and the post-intervention system the “experimental” or “manipulated” system.

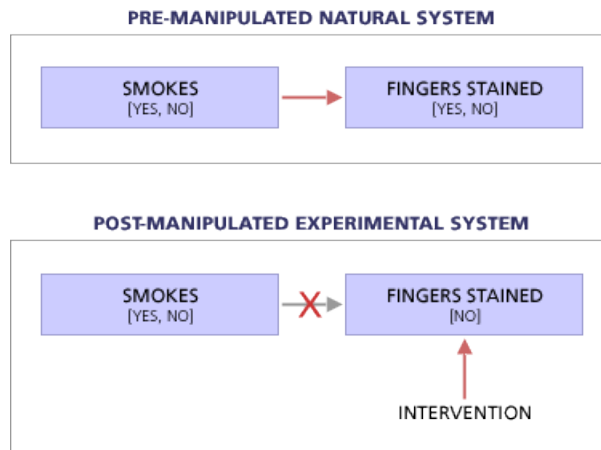
For example, consider smoking and tar stains on your fingers. In the natural course of things, smoking is a cause of tar stained fingers. Thus we draw the following causal graph to represent the “natural” or “pre-manipulated” system:



Consider two ideal interventions, one to ensure that someone smokes, and another to ensure they have no tar stains on their fingers. In the first case, consider the intervention that makes everyone smoke. The post-manipulated system is the same as the pre-manipulated one, because the relationship between smoking and finger stains is unchanged, and it’s the only relationship in the natural system.



In the second case, however, in which we intervene to prevent tar stains, the post-manipulated system is different. Consider an intervention in which we scrub everyone’s hands with tar solvent soap until they are stain free. In that case the influence of smoking on finger stains is broken. It has been usurped by our intervention:



It is important to note here that even though we have been talking about intervening on the entire population – i.e. making everyone smoke, or scrub his or her hands – an intervention is defined as an action performed on a single individual. Thus, we can talk about intervening on a population to make everyone smoke only if we realize what that is: the same intervention on each individual in the population. If we keep this in mind, then we can also talk about intervening on a population, but not making everyone do the same thing. We could, for instance, intervene on the variable SMOKES by making half the population smoke, and preventing the other half from smoking. Or we could intervene on the variable FINGERS STAINED by making half the population scrub their fingers and forcing half the population to rub tar on their fingers. In either case, whether we intervene on FINGERS STAINED by making everyone have the same value or different values, the post-manipulation graph is the same. In both cases we had an ideal intervention, so in both cases the arrow from SMOKES to FINGERS STAINED will be broken.

In general, moving from the pre- to the post-manipulated causal graphs representing the system is simple. To form the post-manipulated graph, begin with the pre-manipulated graph, and:

- Add the intervention as an extra variable(s) to the causal graph with an arrow into the variable(s) it directly changes.
- Break or erase all the arrows pointing into variables that are directly intervened upon.

5.2.5 Module 5 | Exogenous and Endogenous Variables and Causal Structure

It is useful to separate variables in causal systems into two sorts:

- Independent, or exogenous variables, and
- Dependent, or endogenous variables

Independent, or Exogenous Variable A variable is *independent*, or *exogenous*, if it is the effect of no other variable in the system.

Dependent, or Endogenous Variable A variable is *dependent*, or *endogenous*, if it is the effect of at least one other variable in the system.

Because interventions only change the relationship between an effect and its causes, an intervention on an exogenous variable does not change the causal graph at all. Except for adding the intervention variable, the pre-manipulation and post-manipulation causal graphs are the same. We say that such interventions are *structure-preserving*.

Structure Preserving Intervention An intervention is *structure preserving* if it is an intervention on exogenous variables only.

An (ideal) intervention on an endogenous variable does change the relationship between it and its causes, so we call interventions on endogenous variables *structure-altering*.

Structure Altering Intervention An intervention is *structure altering* if it is an intervention on at least one endogenous variable.

5.2.6 Module 5 | Interventions on Cyclic Causal Systems

So far, we have only considered acyclic causal systems, that is, cases in which no variable is a direct or indirect cause of itself. The same rules for changing the causal graph from the pre-manipulated to post-manipulated system apply to cyclic cases, however.

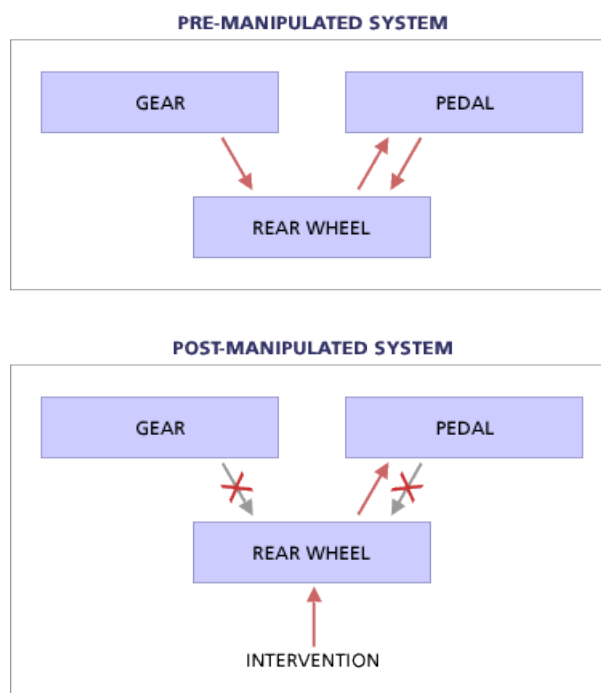
In general, moving from the pre- to the post-manipulated system is simple:

- Add the intervention as an extra variable(s) to the causal graph.
- Break or erase all the arrows pointing into variables that are directly intervened upon.

For example, consider a (simulated) bicycle as a causal system. You may intervene on the system by moving the gear shift, the pedals, or the rear wheel.

In this case, the causal system for the bicycle has no cycles. The rear wheel is turned by pedaling and by setting a gear shift, but the motion of the rear wheel has no causal influence on the motion of the pedals. Contrast this to an older style bicycle in which the pedals and rear wheel are more tightly linked.

In this case the causal relationship between the pedals and the rear wheel is cyclic. If we intervene and change the state of the rear wheel, however, we break the influence of the pedals on the wheel.



5.2.7 Module 5 | Case Studies

As in previous modules, in this section we apply the concepts covered to case studies from the news media.

The first case study describes an attempt by the government in Columbia to reduce violence through the control of guns.

Case Study: Colombian cities experiment with gun ban

Author: Daniel Steel

Date: March 1, 2000

Source : <http://jama.ama-assn.org/>

Copyright: 2001 Carnegie Mellon University

Concepts

- variables

- causal graphs
- interventions
- experiments

Keywords

- guns
- homicide

Intermittent bans on guns proved effective in reducing homicides in the Colombian cities of Bogotá and Cali, an article published in today's issue of the *Journal of the American Medical Association* reports.

Colombia is notorious for being one of the most violent countries on earth, with a homicide rate over four times greater than that of the United States. It is estimated that guns are directly implicated in 80% of all homicides in Colombia.

Faced with this dire crime epidemic, leaders of some Colombian cities have resorted to drastic measures. In 1993, officials of Cali launched the Program for Development, Security and Peace, an important part of which consisted in banning citizens from carrying guns on days in which homicides are most common. These included paydays, holidays, and election days. Bogotá followed suit in 1995.

Colombia has strict gun ownership laws. Legal gun ownership is only possible given a permit issued by the military, who also control the legal sale of firearms. In order to obtain a permit, a citizen is required to demonstrate a genuine need for a gun. However, these regulations are largely undermined by smuggling and a thriving black market.

The ban was enforced by means of police checkpoints at which individuals were searched. Citizens were also searched in the course of ordinary encounters with the police, such as traffic stops.

The researchers found that, controlling for other factors, the homicide rate was significantly lower on those days when the ban was in effect. Cali experienced a 14% drop in homicides, while a 13% reduction occurred in Bogotá.

The lead author of the study, Andrés Villaveces of the Harborview Injury Prevention and Research Center, said that it is uncertain how the program brought about this decrease in homicides. He also cautioned against automatically extrapolating these results to other cities where homicides are less frequent.

The second case describes an attempt to reduce obesity by reducing the hours children spend watching TV and playing video games.

Case Study: For kids, reducing TV viewing may be a key to preventing obesity

Author: Mitch Leslie

Date: May 5, 1999

Source : unknown

Copyright: unknown

Concepts

- variables
- causal graphs
- interventions
- experiments

Keywords

- television
- obesity

A Stanford study suggests that, for grade-school children, watching less television may be a key to limiting weight gain. Children who were involved in a one-year curriculum to reduce their TV viewing gained significantly less body fat than a control group of their peers.

The study of 192 third- and fourth-graders from two San Jose elementary schools is important not only for showing a connection between TV watching and body weight, but it is also some of the first evidence that a behavioral program delivered in schools can motivate children to ration their TV viewing, said the study's author, Thomas Robinson, MD, assistant professor of pediatrics and medicine. Robinson thinks this program is a promising population-wide approach to help prevent childhood obesity, which he says has reached epidemic proportions.

"One of the things that makes this study unique was that it focused specifically on reducing TV, videotape and video game viewing without promoting any other activities as substitutes," Robinson said. "As a result, we were able to isolate the effects of these media alone."

Robinson presented his results May 4 at the Pediatric Academic Societies annual meeting in San Francisco.

American children have become fixated on television and now spend, on average, more than four hours a day watching TV and videos or playing video games, according to nationwide surveys. Only sleep consumes a larger share of the average child's day. And more and more children and adults are now overweight. Rates of childhood obesity have more than doubled over the past two decades, Robinson said.

To physicians and public health experts, this epidemic of obesity is alarming. Overweight children tend to become overweight adults, who are more likely to develop life-threatening health problems like diabetes and heart disease. The amount of time kids spend immobile in front of the TV has frequently been blamed as a cause of obesity. That proposal sounds eminently logical, Robinson said, but epidemiological studies seeking a link between number of hours watched and weight have found only a weak relationship, he noted.

All of these studies have looked for correlations between the TV viewing and body weight without manipulating the amount of TV watched. Robinson approached the problem differently by trying to alter TV habits and then subsequently measuring the effect on body weight. As a result, this was the first experimental test of the TV-obesity hypothesis in normal school children.

Local education officials picked two schools with similar ethnic composition, socioeconomic standing and scholastic achievement. At one of the schools, the third- and fourth-graders received an 18-lesson program, presented by their classroom teachers as part of the normal school curriculum, that was designed to reduce TV and videotape watching and video game playing. Both schools agreed to participate before learning which school would receive the curriculum, and the students at each school were found to have similar TV viewing habits and body fatness at the beginning of the school year, Robinson said.

TV-reduction lessons began by encouraging children to keep track of how much time they spent watching TV and videos and playing video games. In the next step, the children were encouraged to go 10 days without TV – something that two-thirds of them accomplished. Then the lessons turned to setting TV-watching limits of 7 hours per week and becoming more selective viewers.

At the beginning and the end of the school year, pupils at both schools filled out surveys about their media use, diet and exercise habits. Their body fat was measured, and they took a physical fitness test. Parents also completed interviews about their own and their children's habits.

To account for normal growth and maturation, Robinson compared the change in different body measurements between the two schools over the course of the school year. Children attending the school that had received the TV-reduction lessons did reduce their television watching by about one-third to one-fourth, compared with their peers. Over the course of the study, they also showed a significantly smaller increase in waist size, waist-to-hip ratio and body mass index, a measure of weight adjusted for height. For example, the average pupil at the TV-reduction school gained nearly 2 pounds less than the average student at the control school.

However, the two schools didn't differ in consumption of high-fat foods, amount of moderate-to-vigorous exercise, or physical fitness. So if the kids weren't exercising more or eating a healthier diet, Robinson needed to try to account for the weight difference. Robinson said one possibility was that the kids were performing more low-level activities – more energetic than simply sitting still but less energetic than walking. Freed from the TV, the children may simply have been moving around more, though this kind of activity is extremely difficult to measure, Robinson said.

However, the difference may also stem from changes in the number of meals the kids ate in front of the TV. The children at the TV-reduction school significantly decreased the number of meals they ate while watching TV. Even a small change in caloric intake could account for the difference in weight gain, Robinson noted. For example,

the calorie equivalent of one additional regular soft drink per day translates into a yearly weight gain of about 15 pounds, he said.

“These are the largest effects on body composition we have seen from any obesity prevention program we have studied to date,” said Robinson. “But our enthusiasm has to be tempered because the results come from only a two-school study.”

Robinson’s study was funded by a grant from the American Heart Association, California Affiliate. He subsequently has received a grant from the National Heart, Lung and Blood Institute to replicate the study with 900 students from 12 elementary schools. That research has just begun and will also involve Stanford researchers Joel D. Killen, PhD, associate professor of medicine; Helena C. Kraemer, PhD, professor of psychiatry and behavioral sciences; William Haskell, PhD, professor of medicine; Leslie Pruitt, PhD, research associate in medicine; and Donna Matheson, PhD, a post-doctoral scholar in pediatrics.

5.3 Summary

5.3.1 Module 5 | Module Summary

The module stresses six basic skills and/or concepts:

5.3.2 Concepts and Skills Covered in Module 5

1. Intervention as Action
2. Ideal Interventions
3. Interventions and Causal Relationships
4. Pre-Manipulation vs. Post-Manipulation Graphs
5. Exogenous vs. Endogenous Variables
6. Learning Causal Effects from Interventions

Causal knowledge informs us about what would happen if we intervened to act in certain ways. The simplest kinds of interventions, which we call *ideal interventions*, are clearly defined actions from outside the causal system that directly determine (fix) the value of one variable in the causal system. In this module we have only covered ideal interventions.

Ideal interventions not only change the value of a variable in a causal system, they sometimes change the causal relationships that obtain in a causal system. Interventions that change the value of a variable that is a cause in the system do not change the relationship between the cause and its effects. Interventions that change the value of an effect do change the relationship between the effect and its causes; they break the relationship between the effect and its causes – because, by setting the value of the effect, the intervention prevents the causes from having any influence on the effect.

The causal graph that describes the relations among the variables in a causal system before we intervene on it is called the *pre-manipulation* graph, or the natural system. The causal graph that describes the causal relations that obtain among the variables in the same system after an intervention is called the *post-manipulation*, or experimental graph.

A variable that is only a cause in a system is called *exogenous*, or independent. A variable that is an effect of at least one other variable in the system is called *endogenous*, or dependent. Interventions that change the value of an exogenous variable are said to be structure-preserving, and interventions that change the value of an endogenous variable are said to be structure-altering. If an intervention is structure-preserving, the pre-manipulation and post-manipulation causal graphs are the same. If an intervention is structure-altering, the pre-manipulation and post-manipulation graphs are different.